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Review Article

A review of hydrogen production and supply chain modeling and optimization

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HIGHLIGHTS

- Recent optimization models for hydrogen supply chains and production are reviewed.
- The multiple components of hydrogen supply chains are discussed.
- The main features of the models and solution methods used are analyzed.
- Opportunities in modeling are identified and future research directions suggested.

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ABSTRACT

This paper reviews recent optimization models for hydrogen supply chains and production. Optimization is a central component of systematic methodologies to support hydrogen expansion. Hydrogen production is expected to evolve in the coming years to help replace fossil fuels; these high expectations arise from the potential to produce low-carbon hydrogen via electrolysis using electricity generated by renewable sources. However, hydrogen is currently mainly used in refinery and industrial operations; therefore, physical infrastructures for transmission, distribution, integration with other energy systems, and efficient hydrogen production processes are lacking. Given the potential of hydrogen, the greenfield state of infrastructures, and the variability of renewable sources, systematic methodologies are needed to reach competitive hydrogen prices, and design hydrogen supply chains. Future research topics are identified: 1) improved hydrogen demand projections, 2) integrated sector modeling, 3) improving temporal and spatial resolutions, 4) accounting for climate change, 5) new methods to address sophisticated models.
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Introduction

The use of hydrogen as an energy carrier is by no means new. Hydrogen has been a topic of discussion to fuel the transportation sector since the 1970s. However, due to the exorbitant investment costs, high production costs, and availability of cheaper fuel alternatives, the promise of hydrogen fuel has not come to fruition [1]. Other concerns surrounding the safety of containing hydrogen have also made its adoption difficult. Nevertheless, the growing need for countries to reduce greenhouse gas emissions and seek cleaner fuels has allowed hydrogen to reemerge as a potential energy carrier that could help to decarbonize various sectors [2–4]. The focus of hydrogen is no longer on the transportation sector alone, but also on industries such as heating, chemical, and steel, where reducing emissions has historically been difficult. Low-carbon hydrogen can be used as a substitute for natural gas to chemically reduce iron ore in the steel industry. Direct reduction of iron with hydrogen would decrease the overall thermal energy requirement [5]. In the chemical sector, electrolyzer and carbon capture and storage (CCS) technologies, have been suggested to help reduce emissions related to ammonia production by replacing hydrogen from reforming natural gas with cleaner alternatives [6].

Hydrogen may also enhance the sustainability, reliability, and flexibility of energy systems. Hydrogen can complement the integration of renewable technologies in the power sector, allowing surplus renewable energy to be stored and utilized later [2]. Similarly, hydrogen can be produced in regions with high renewable energy potential and transported long distances to countries with high energy demands and limited energy resources [7]. In the gas sector, hydrogen can be blended with natural gas to produce cleaner combustible fuels. As of 2021, 17 countries, including Japan [8], Germany, and South Korea, have begun preparing hydrogen strategies for the gradual rollout of new applications, infrastructure, technical and institutional hydrogen production, and trade rules [9].

Today, roughly 95% of all hydrogen is produced from fossil fuels through the steam methane reformation (SMR) of natural gas [3,10]. There is currently no significant production of

hydrogen from renewable sources. The scarcity of green hydrogen generation facilities and hydrogen storage and transportation infrastructures are the primary barriers to establishing and growing a hydrogen economy. However, numerous green hydrogen pilot projects are rolling out, which will shift production towards more renewable methods [9]. There have also been significant advances in hydrogen technologies such as electrolyzers and hydrogen fuel cells; investment costs have decreased over the years while technological efficiencies have increased [6].

In 2019, the International Agency for Renewable Energy (IRENA) released a report in response to a call in the G20 Karuizawa Energy Innovation Action Plan to develop an extensive analysis of the long-term strategies and pathways necessary to create a clean hydrogen economy. IRENA indicated that hydrogen would cover 6% of the world's total energy demand [3]. In contrast, the Hydrogen Council (formed by a group of companies) estimated that nearly 18% of the world's energy demand could be met with hydrogen [4].

From an economic perspective, comprehensive investment strategies [11–17] are required for research to bring low-carbon hydrogen-related technologies to maturity and promote the production, distribution, and consumption of hydrogen across various sectors. Ultimately, the goal is to bring the cost of low-carbon hydrogen to competitive values—\$1/kg H_2 in one decade—as targeted by the U.S. Department of Energy [18]. Given the potential role of hydrogen across multiple sectors, hydrogen has the potential to disrupt the energy sector and the overall economy. The energy sector and the economy are highly interdependent with energy deemed the basis of industrialized economies [19]. Therefore, as hydrogen penetrates as a new energy vector and exhibits a relevant role in power systems, transportation, and industrial processes, it may influence the prices of competing energy vectors and economies. The 1970s were marked by a surge in world oil prices and a resulting period of stagflation—stagnant economic growth and high inflation rates. Similarly, current energy price shocks have contributed to rampant inflation and slowed economic growth enough that countries fear a recession [20]. Noting the impact that the energy sector has on global economics, it is vital to assess the effects that expanding the hydrogen supply chain may have on the economy of hydrogen-producing and consuming countries.

Two fundamental research topics required to address the development of the hydrogen economy are the design of the hydrogen supply chain and the process design of a hydrogen production facility (or small-scale/self-contained network). Hydrogen supply chain design (HSCD) addresses issues related to the deployment of hydrogen infrastructure and considers the feedstocks used in the production, the production itself, and the storage, transportation, distribution, and end-uses of hydrogen. A depiction of a hydrogen supply chain superstructure is shown in Fig. 1. In the context of the hydrogen supply chain, a superstructure is a network representation that includes multiple alternatives for feedstock, production, storage, transportation, and distribution. The development of an efficient and reliable hydrogen supply chain requires a deep understanding of the system network, the tradeoffs between technologies, and the availability of local resources [21]. The optimization of superstructures considers the interaction between various parts of a supply chain. Optimization enables the simultaneous selection of the optimal network (from the superstructure) and the optimal operating conditions.

Hydrogen is incredibly versatile and can be produced through electrolysis of water, gasification of biomass and coal, and SMR of natural gas or continuous catalytic reforming (CCR) of petroleum-based naphthas. Other production methods via the catalytic decomposition of natural gas [22], and biological processes by microbes and using nanoparticles [23], are being explored.

Hydrogen can be transported to terminals via trucks, trains, pipelines, or ships. Currently, most hydrogen consumed is obtained locally because long-distance hydrogen transport is not economical due to its low volumetric energy density [10]. At terminals, hydrogen is stored in liquid (cryogenic) or gaseous (compressed) phases until it is distributed to end-users. The end-users can be power plants, refueling stations, chemical plants (to produce ammonia or e-fuels), or refineries, to name a few.

Hydrogen process design (HPD) focuses on a specific section of the hydrogen supply chain, the production facilities. HPD details the design of new facilities in consideration of the product output, expected process yields, and throughput rates. The design begins with the process synthesis (the choice of technologies and industrial components required) considering budgetary constraints, capital and operating costs of a process site, space availability, safety measures, and environmental impact. HPD considers localized production and consumption of hydrogen, such as hydrogen usage within refineries or small-scale stand-alone networks. The design of a complete supply chain for hydrogen naturally includes the design of the hydrogen production process. However, the design of the production is treated with a different level of detail when compared to the design of the production process. At the HSCD level the problems tend to have longer time horizons and multiple components, multiple decision levels, and therefore, coarse time resolutions and lower levels of detail in the description of the components' performance are involved. On the other hand, HPD problems focus on less components (no transportation, single site, no choice between processes or feedstocks) and address the production component in greater detail. Therefore, despite the overlap between HPD and HSCD problems, we reviewed them as two distinct problems.

Motivation

Given the potential of hydrogen in the energy transition, the greenfield state of infrastructures, and the variability of renewable sources, systematic methodologies are needed to reach competitive low-carbon hydrogen prices, and design and plan complex integrated hydrogen supply chains. This study aims to critically analyze the systematic methods and models proposed to optimize hydrogen supply chains and production processes and to identify missing features, coarse levels of detail, and identify new research directions.

Reviews exploring hydrogen supply chain design and production already exist, but to the best of our knowledge, no comprehensive review of both has yet been performed. For

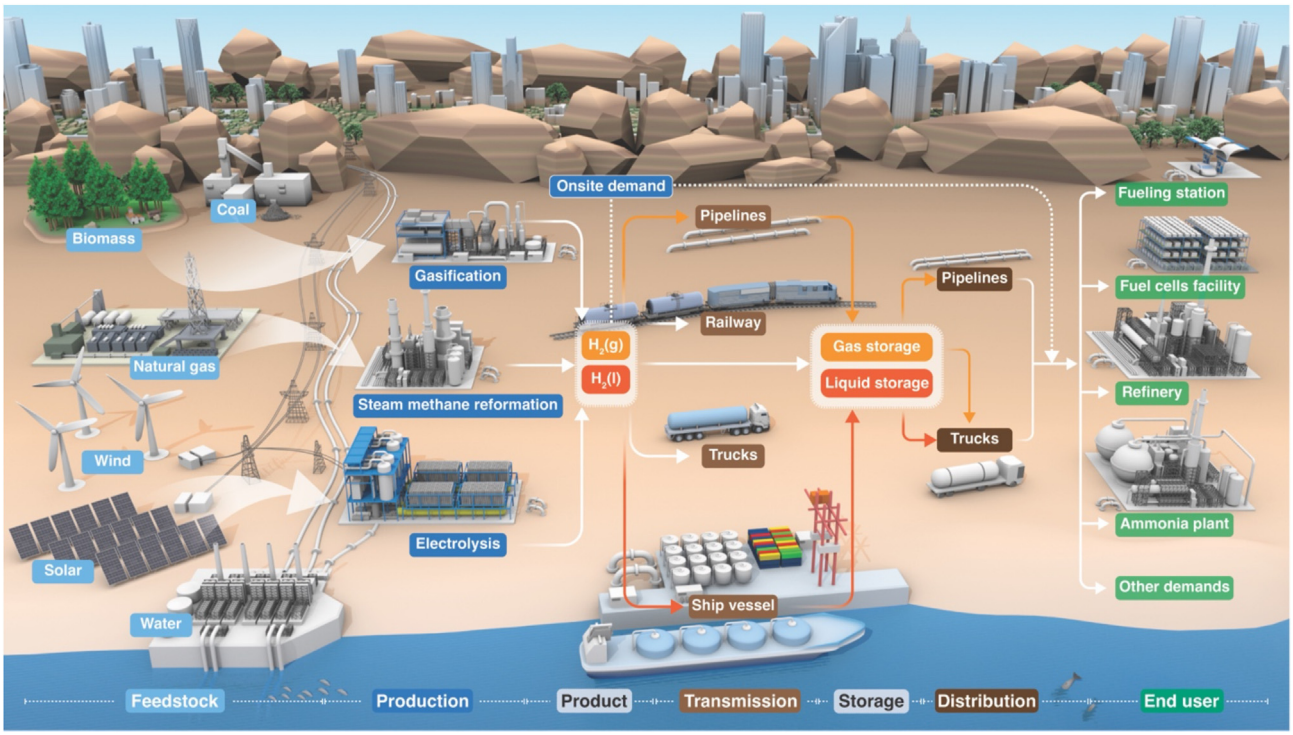


Fig. 1 – Overview of a hydrogen supply chain superstructure. Hydrogen can be produced and utilized onsite, stored, and distributed locally (via pipeline as gas or trucks as liquid or gas), or transported long distances (via pipelines as gas, railways, and trucks as liquid or gas, or via shipping vessels as liquid).

example, Bolat and Thiel [24,25] perform a two-part review of the hydrogen supply chain, presenting 1) the architecture of the hydrogen supply chain pathways and 2) the techno-economic inputs for hydrogen production. In their first paper [24], the authors highlight a superstructure of the hydrogen supply chain and describe how hydrogen could theoretically get from the production site to the end-user, the hydrogen fuel cell vehicles (HFCV). In the second paper [25], the authors present the technical specifications of hydrogen production technologies, such as the availability factor, capacity, feedstock inputs, and plant lifetimes for gasification, SMR, and electrolytic-based hydrogen production. Moreover, they provide the capital cost and fixed and variable operating costs for each process. Overall, these reviews [24,25] give a good overview of the structure of the hydrogen supply chain. However, the authors did not consider current hydrogen supply chains, such as refining and fertilizer production, or explore how these supply chains can play a role in the hydrogen economy. Similarly, the authors did not consider additional model inputs such as hydrogen demand projections or methods to calculate demand.

Maryam [26] reviews the modeling techniques for deploying a hydrogen supply chain in the UK. The author addresses key components of optimization models, emphasizing the periodicity of the models (single vs. multi-period approaches) and the objectives that are optimized (single vs. multi-objective strategies). Financial costs and environmental objectives are discussed, but additional performance measures are not explored. Similarly, uncertainty metrics in HSCs are briefly mentioned but not categorized and solution methods are not discussed.

Lui and Ma [27] assess the hydrogen supply chain problem by performing a component analysis, breaking down the system into the production, terminals, storage, transportation, and fueling stations. The authors mention three performance measures for optimization-based approaches: cost, safety, and carbon externalities. For costs, they specify capital and operating costs for hydrogen production and storage facilities, as well as the costs of establishing and managing a hydrogen transportation network. However, the authors do not consider financial risk metrics or explore the profits that may result from HSC. While Lui and Ma [27] explore the approaches used to solve HSCD problems, categorizing them as exact methods, heuristics, and simulations, they do not distinguish between the solution methods for problems that deal with uncertainty (vs. deterministic) or that are multi-objective (vs. single objective).

This review examines the technologies presented along the hydrogen supply chain (or production stage) and the inputs necessary to produce the fuel. The paper also assesses the assumptions used in the literature to determine hydrogen demand (transport, energy storage, heating, refining, and e-fuel production); a topic that has not been highlighted in previous review papers. Moreover, a detailed classification of the HSCD and HPD models found in the literature is performed. We account for the type of models used, whether uncertainties are considered and how they are handled, and the performance measures considered in the models (assessment of cost, safety and risk, and environmental concerns).

We identify key features neglected in the current HSCD and HPD models found in the literature and provide

recommendations to build more realistic models. The latter pertain to various areas including 1) improved hydrogen demand projections, 2) integrated sector modeling, 3) improving temporal and spatial model resolutions, 4) accounting for climate impacts on a developing hydrogen economy, and 5) new methods to solve more complex hydrogen supply chain and production models.

The present paper is organized as follows. Section Classification of HSCD and HPD Models describes the methodology used to gather and classify the research papers. In Section Supply Chain Component Analysis, a detailed supply chain component analysis is presented.

Modeling approaches and solution methods are discussed in Sections Modeling Features and Solution Methods, respectively. Section Summary and Recommendations provides the concluding remarks, together with a brief discussion of alternative research directions and potential problems to explore in the field of HSCD and HPD.

Classification of HSCD and HPD models

An extensive search of the peer-reviewed literature was implemented using keyword queries of the primary electronic bibliographical sources Google Scholar, Scopus, and Web of Science. Keywords included hydrogen, supply chain, network design, economic analysis, optimization, process design, and production optimization. Between 2000 and 2022, we see a growing trend in the number of hydrogen-related publications, most of which appeared in the *International Journal of Hydrogen Energy*. A total of 99 papers were identified, of which 85 focused on HSCD and 14 covered HPD.

Fig. 2 shows the geographical distribution of the case studies used HSCD and HPD models. South Korea was most frequently (18 times) used as a case study, followed by the United States France (11), and the UK (10). Eastern Europe, North America, and East Asia are the most represented regions. Only a small number of papers explore the Middle East (3: Qatar and Iran), Southeast Asia (6: Malaysia, Indonesia, Singapore), South Asia (1: Pakistan), Oceania (1: Australia), and Africa (1: South Africa).

The models were further classified by model type, use of uncertainty, spatial scale, and the periodicity of investment decisions. The characteristics of the models are summarized in Table 1.

HCSD problems are predominately set up as mixed-integer linear programs (MILP) models. Six papers [44,55,69,100,113,114] formulated the model as a linear

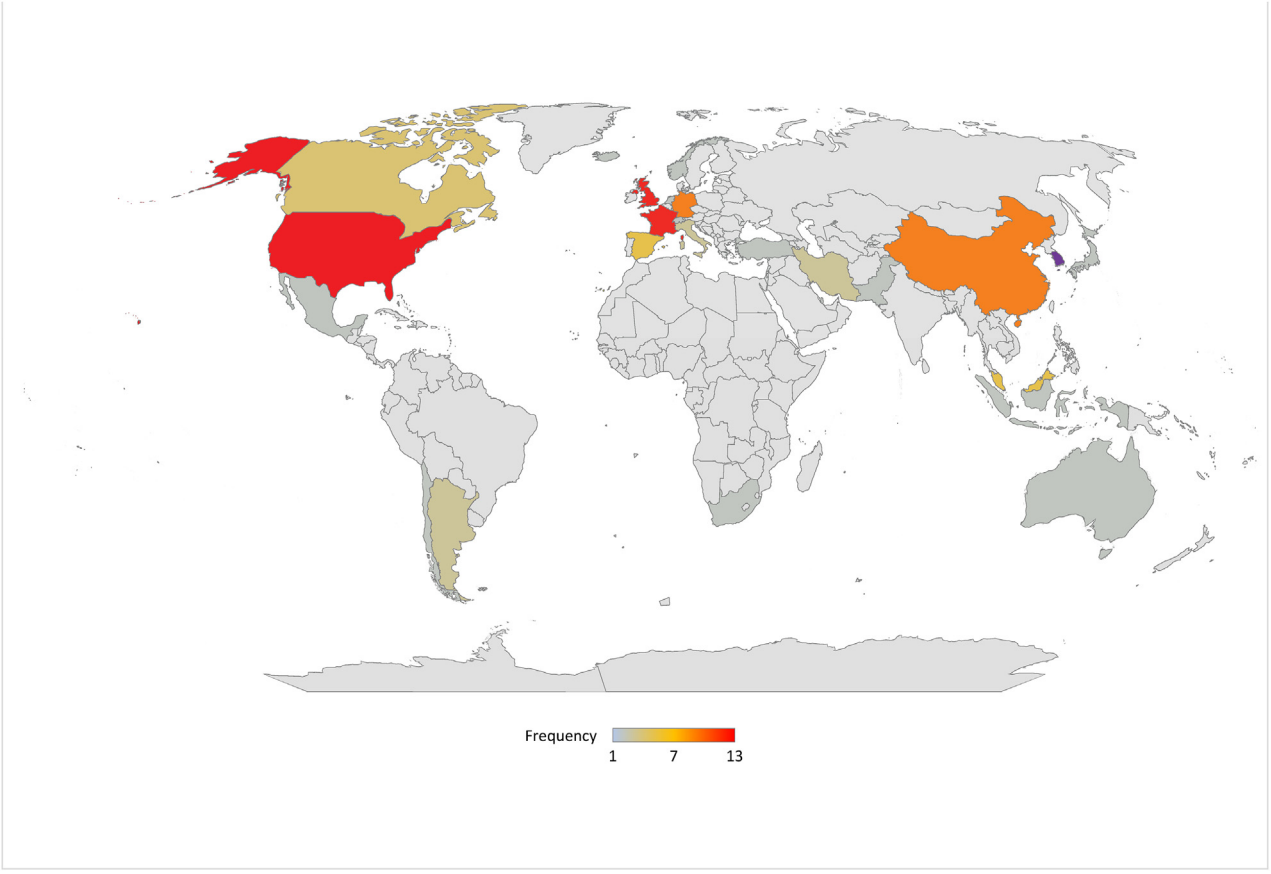


Fig. 2 – Countries used as case studies. Most case studies take an Eastern European, East Asian or North American perspective. Few papers use the Middle East or South America as cases studies. South Korea (in purple) is used as a case study 18 times; for scaling purposes its value is not included in the color bar.

Table 1 – Characterization of the HSCND and HPD modeling and optimization approaches found in the literature.

Reference	Model type							Uncertainty	Spatial scale					Investment Stages	
	LP	NLP	MILP	MINLP	SIM	TEA	OTH		I	N	R	U	P	Single	Multi
Agnolucci et al. (2013) [28]			•							•					•
Almansoori & Betancourt-Torcat (2016) [29]			•							•				•	
Almansoori & Shah (2006) [30]			•							•				•	
Almansoori & Shah (2009) [31]			•							•					•
Almansoori & Shah (2012) [32]			•					•		•					•
Andre et al. (2013) [33]		•								•				•	
Andre et al. (2014) [34]		•									•				•
Armijo et al. (2020) [35]						•			•					•	
Bae et al. (2020) [36]			•									•			•
Balaji et al. (2020) [37]					•								•	•	
Carrera & Azzaro-Pantel (2021) [38]			•								•				•
Carrera & Azzaro-Pantel (2021) [39]			•								•				•
Cerniauskas et al. (2019) [40]							•			•					•
Chen et al. (2020) [41]			•				•						•	•	
Cho & Kim (2019) [42]			•							•				•	
Cho et al. (2016) [43]			•								•				•
Contaldi et al. (2008) [44]	•									•					•
Coppitters et al. (2019) [45]				•				•					•	•	
Dagdougui et al. (2012) [46]			•								•			•	
Dayhim et al. (2014) [47]			•					•			•				•
De-León Almaraz et al. (2013) [48]			•							•				•	
De-León Almaraz et al. (2014) [49]			•								•				•
De-León Almaraz et al. (2015) [50]			•							•	•				•
Demirihan et al. (2020) [51]			•								•			•	
Demirihan et al. (2021) [52]			•								•			•	
Fazli-Khalaf et al. (2020) [53]			•					•		•					•
Gibson & Kelly (2008) [54]							•						•	•	
Gim et al. (2012) [55]	•									•				•	
Gondal & Sahir (2012) [56]				•						•				•	
Guillen-Gosalbez et al. (2010) [57]			•							•					•
Guler & Erdogan (2021) [58]			•							•					•
Guo et al. (2022) [59]			•				•						•	•	
Han et al. (2013) [60]			•							•				•	
He et al. (2017) [61]				•							•	•		•	
Heuser et al. (2019) [7]						•			•					•	
Hong et al. (2021) [62]						•			•					•	
Hugo et al. (2005) [63]			•												•
Hwangbo et al. (2018) [64]			•					•		•				•	
Hwangbo et al. (2016) [65]				•				•					•	•	
Hwangbo et al. (2017) [66]			•					•		•				•	
Ingason et al. (2008) [67]			•							•				•	
Jagannath & Almansoori (2014) [68]				•				•					•	•	
Jiang et al. (2022) [69]	•										•			•	
Joffe & Bauen (2004) [70]												•		•	
Johnson & Ogden (2012) [71]			•								•			•	
Kamarudin et al. (2009) [72]			•								•			•	
Khalilnejad et al. (2016) [73]				•									•	•	
Kazi et al. (2021) [74]			•										•	•	
Kim & Kim (2016) [75]			•							•				•	
Kim & Kim (2017) [76]			•								•			•	
Kim & Moon (2008) [77]			•					•		•				•	
Kim et al. (2008) [78]			•					•		•				•	
Li et al. (2008) [79]			•							•					
Li et al. (2018) [80]															
Li et al. (2021) [81]			•							•					•
Li et al. (2020) [82]			•								•			•	
Lim et al. (2020) [83]										•				•	
Lou et al. (2014) [84]				•				•					•		
Mah et al. (2020) [85]			•								•			•	
Mah et al. (2022) [86]			•								•			•	
Martin & Grossman (2018) [87]			•					•		•	•			•	
Moreno-Benito et al. (2017) [88]			•							•					•

Table 1 – (*continued*)

Reference	Model type							Uncertainty	Spatial scale					Investment Stages	
	LP	NLP	MILP	MINLP	SIM	TEA	OTH		I	N	R	U	P	Single	Multi
Murthy Konda et al. (2011) [89]			•							•					•
Nunes et al. (2015) [90]								•		•					
Ochoa Bique & Zondervan (2018) [91]			•							•				•	
Ochoa Bique et al. (2019) [92]			•							•					
Ochoa Robles et al. (2019) [93]			•					•			•				•
Ochoa Robles et al. (2020) [94]			•								•				•
Ochoa Robles et al. (2018) [95]			•					•			•				•
Ogumerem et al. (2018) [96]			•								•				•
Ogumerem et al. (2019) [97]			•								•				•
Parker et al. (2010) [98]				•							•			•	
Qadrdan et al. (2008) [99]							•			•					•
Reuß et al. (2019) [100]	•			•							•			•	
Reyes-Barquet et al. (2022) [101]			•							•					•
Sabio et al. (2010) [102]			•					•		•					•
Sabio et al. (2012) [103]			•							•					•
Samsatli et al. (2016) [104]			•							•					•
Sardashti Birjandi et al. (2014) [105]		•		•									•	•	
Saur & Ramsden (2011) [106]						•							•	•	
Schnuelle et al. (2020) [107]					•	•							•	•	
Seo et al. (2020) [108]			•							•				•	
Shamsi et al. (2019) [109]			•										•		•
Shamsi et al. (2021) [110]			•								•			•	
Stephens-Romero et al. (2010) [111]			•									•			•
Stiller et al. (2010) [112]										•					•
Stöckl et al. (2021) [113]	•									•				•	
Strachan et al. (2009) [114]	•									•				•	
Sun et al. (2017) [115]		•										•		•	
Talebian et al. (2019) [116]			•								•				•
Tlili et al. (2020) [117]						•				•				•	
Weber & Papageorgiou (2018) [118]			•							•				•	
Won et al. (2017) [119]			•								•			•	
Woo & Kim (2019) [120]				•							•			•	
Woo et al. (2016) [121]			•								•			•	
Yáñez et al. (2018) [122]			•								•				•
Yang & Ogden (2013) [123]			•								•				•
Yang et al. (2020) [124]			•					•			•			•	
Yoon et al. (2022) [125]			•							•					•

LP: Linear program: MILP: Mixed-integer linear program; MINLP: Mixed-integer nonlinear program; SIM: simulation; TEA: techno-economic analysis; OTH: dynamic programming, model predictive control, laboratory experiments; I: International; N: National; R: Regional; U: Urban; P: Plant.

program (LP). Three papers [33,34,115] use a nonlinear programming (NLP) model and five [56,61,98,100,120] use a mixed-integer nonlinear programming (MINLP) approach. Four papers [7,35,62,117] perform a techno-economic analysis (TEA). One paper [99] uses dynamic programming to solve the HSCD problem. HPD problems, on the other hand, are set up as simulations (SIM) [37,107], MILP [41,59,74,109], NLP [105] or MINLP [45,65,68,73,84,105] formulations, laboratory experiments [54], or model predictive control (MPC) [41] problems. Two papers [106,107] performed a techno-economic analysis of the HPD problem.

Regarding uncertainty, 17 papers incorporate uncertainty into their analysis. A breakdown of the uncertainties explored and the method of integrating uncertainty into their analysis is examined in the section entitled Uncertainty below.

In terms of the spatial scale, national and regional HSCD problems dominate the literature. Few papers explore the hydrogen supply chain at an international level (3) or within an urban area (5). Note that understanding how the hydrogen supply chain may be structured and developed between countries is essential as some countries begin discussions and plan to produce hydrogen primarily for export purposes. Countries with high renewable potential, such as Saudi Arabia [126,127], Oman, Egypt, and Chile [7], have already announced projects to export hydrogen to nations that foresee a significant demand for cleaner fuel alternatives. Countries such as Japan, South Korea, and China are expected to have enormous demands for hydrogen by 2050. Since the shutdown of all its nuclear facilities following the Fukushima Daiichi nuclear disaster, Japan relies primarily on fossil fuel imports for power generation [7]. The Japanese government has stated its need to establish a sustainable hydrogen economy to reduce greenhouse emissions [8]. Therefore, Japan would require a sustainable and reliable international hydrogen supply chain.

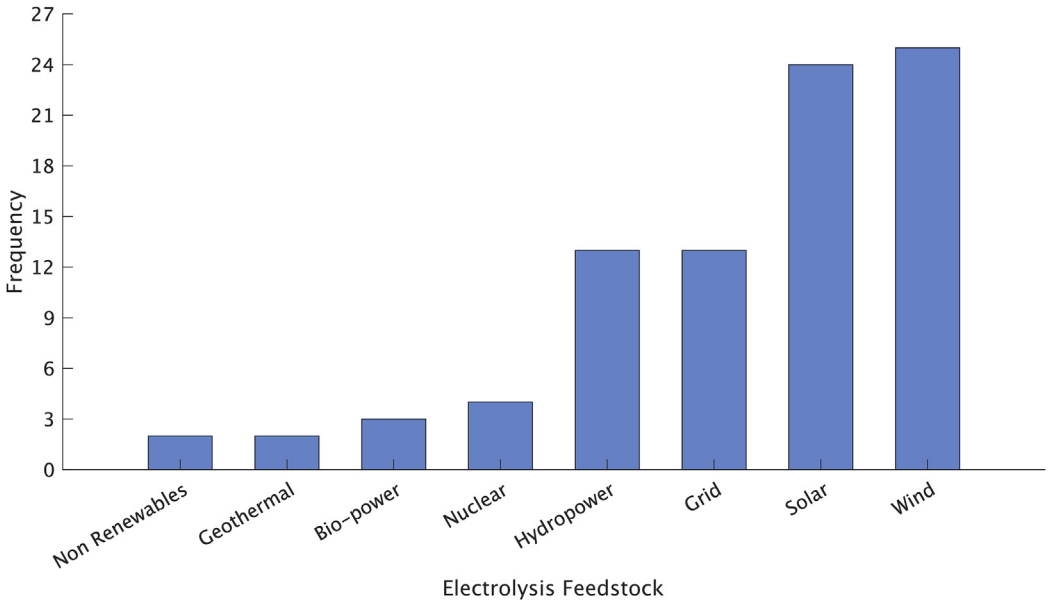


Fig. 3 – Electricity sources for electrolysis. Wind and solar were most commonly used to generate electricity feedstocks for electrolysis. Geothermal was used the least for electrolytic purposes.

HPD problems deal with the production side of the hydrogen supply chain, where the focus is on the plant design to meet a specific hydrogen demand or maintain constant production. The spatial scale is generally the plant or a local site.

Supply chain component analysis

Feedstocks

Hydrogen is an energy carrier that can be produced from numerous feedstocks given the appropriate chemical reaction schemes. The most common feedstocks used in hydrogen production are natural gas, biomass, coal, water, and electricity. SMR and pyrolysis use natural gas and steam to produce hydrogen, while gasification uses oxygen and coal or biomass; electrolysis requires water and electricity.

Feedstock availability is fundamental to hydrogen-related problems. Some countries, such as Norway, have abundant natural gas reserves and therefore may consider it for hydrogen production; on the other hand, other countries, such as Chile, have abundant wind resources. For intermittent resources such as wind and solar, availability plays a crucial role in hydrogen production. For biomass, considering its seasonal availability is necessary to understand the hydrogen production capacity throughout the year [42,43,75,121]. As the momentum for cleaner fuel alternatives increases, renewable sources have received increasing attention as a means to produce hydrogen. Renewable-powered electrolysis is viewed as a key production method to help decarbonize numerous sectors. In the literature, electricity for electrolytic purposes can be obtained from stand-alone renewable sources or directly from the power grid. Solar and wind power are the primary renewable sources explored in the literature, as depicted in Fig. 3. Hydropower for electrolytic systems was applied to case studies in France, Turkey [58], Iceland [67], China [69], the US [96], and Canada [116]. Nuclear power was found in four references using France as a case study. As France derives more than 75% of its electricity from nuclear, it is expected that nuclear will also be used to power electrolytic processes. Geothermal-based electrolysis was found in two case studies with hydrogen supply chains in Iceland [67] and Turkey [58].

Table 2 – [a]Feedstock requirements and LCOH for different production methods.

Input	Units	SMR	SMR + CSS	CG	CG + CCS	BG	BG + CCS	Nuclear Elect	PV Elect.	Wind Elect	Pyrolysis
Natural Gas	kg	3.36	3.76								4.86
Electricity	kWh	0.31	1.11		1.36		3.68	54.2	54.2	54.2	
Coal	kg			8.51	9.7						
Biomass	kg					36.28	36.34				
Water	kg	21.9	23.7	11.28	38.08	47.48	47.96	10	10	10	8.08
LCOH	2019$/kg	1.35	2.01	1.38	2.32	2.4	3.71	4.3	9.49	5.61	1.87

SMR: steam methane reformation; CG: coal gasification; BG: biomass gasification; CCS: carbon capture and storage; Elect: Electrolysis.
[a] Table adapted from Al-Qahtani et al. [129].

The cost of hydrogen production via electrolysis is heavily dependent on the renewable resource input and its availability. Parkinson et al. [128] and Al-Qahtani et al. [129] quantified the levelized cost of hydrogen (LCOH) from nuclear, wind, and PV solar-powered electrolysis (in a PEM system). They found that PV solar power electrolysis resulted in the highest LCOH compared to nuclear and wind electrolysis as shown in Table 2.

Feedstock purchasing costs are typically considered in the literature, but they are often exogenous inputs. Exogenous feedstock costs prevent hydrogen production/supply chain models from seeing the dynamics of other supply chains and reacting accordingly. Disruptions in supply chains may result in feedstock shortages and subsequent price shocks. This may directly impact the cost of hydrogen production. Parkinson et al. [128] presented the LCOH values for pyrolysis under varying natural gas prices found in the literature. High natural gas price inputs (~$7/GJ) resulted in LCOH values of 2016 $2.5–3.5 kg/compared to the 2016 $1–1.5/kg H_2 when natural gas prices were ~2016 $3.25/GJ. They found similar results in the literature for SMR; natural gas prices between 2016 $2.5–3.5/GJ resulted in LCOH values between $1-$1.25/kg H_2 while natural gas higher prices over $7/GJ led to LCOH values around $1.60/kg H_2 and above. Similar results were found for biomass and coal gasification, with increasing feedstock costs resulting in higher LCOH values. Saur and Ramsden [106] analyzed the system configuration for large-scale wind-powered electrolytic hydrogen production using the H2A [130] model by NREL. They found that the cost of wind electricity greatly impacts hydrogen cost, with higher wind costs resulting in higher hydrogen production costs. They also found that the capacity factor of the wind turbines, i.e., the ratio of generated wind power to the turbine capacity, is correlated with hydrogen costs. Specifically, higher capacity factors are associated with lower hydrogen costs. The authors calculated an LCOH of $4/kg H_2.

A key feedstock that is neglected in the literature is water for hydrogen production. All production methods require some form of water input. Table 2 indicates that biomass gasification requires the largest water input, 47.48 kg of water per 1 kg of hydrogen produced, while pyrolysis requires the smallest water input at 8.08 kg of water per 1 kg of hydrogen produced [129]. In the literature we reviewed, water input flows are explicitly considered for electrolysis, however, they are largely neglected for other technologies.

Ogumerem et al. [96] consider process water material costs for electrolysis in Texas. However, they assumed water process capacities to be unlimited. Schnuelle et al. [107] also considered the cost of purchasing deionized water but again did not account for water availability.

Fazli et al. [53] considered green hydrogen production to decarbonize the industrial sector of Qatar. Water resources were obtained directly from the government for a cost or from industrial processes that produce wastewater byproducts. The wastewater was treated and cycled to electrolytic production facilities. The amount of wastewater that could be treated was limited to the output from the industrial process.

Won et al. [119] modeled a hydrogen supply chain for South Korea and considered the desalination of seawater as an input for electrolysis. In this regard, seawater is abundant, however, the amount of hydrogen that can be produced is limited by the amount of water that can be processed by the desalination plants.

Water is a significant factor in the hydrogen supply chain in regions with limited water availability. MENA countries such as Saudi Arabia [126,127] and Oman have begun rolling out hydrogen and ammonia production projects. A recent study by Riera et al. [131] proposed a novel co-optimization model integrating water desalination and renewable power systems for Saudi Arabia. They show that co-optimizing the two sectors can help reduce total system cost and it enables added flexibility to the sectors as they are designed to respond to one another. With hydrogen as a prospect, it is crucial to develop integrated models that consider the hydrogen supply chain and water supply chain (e.g., installed capacity cost, operating costs, water price) to meet hydrogen production targets.

For gasification, oxygen is a necessary feedstock that helps react carbonaceous feedstocks into gases at high temperatures without combustion. For this process, quantifying the amount of oxygen required and the associated purchase costs would be necessary for hydrogen production. Currently, no paper we reviewed addresses the cost of oxygen. However, two papers [75,119] quantified the amount of oxygen produced as a byproduct via water electrolysis. That byproduct oxygen could then be stored and used as a feedstock for gasification. Won et al. [119] noted that the oxygen flow rate as a byproduct of electrolysis is high during the winter because biomass gasification is less active during that season (limitations of biomass feedstock availability).

Quantifying feedstock requirements, feedstock costs, and feedstock availability are crucial to obtaining an optimal configuration of a hydrogen-based network. By neglecting some feedstock costs, some production methods may appear artificially cheaper than they are. In the same respect, neglecting resource availability may cause a model to select a production process that is not well suited to a region of interest.

Hydrogen production

In the literature we reviewed, hydrogen production was modeled through various technologies and processes. However, the dominating technologies include SMR, electrolysis, and gasification, each with significant tradeoffs. Recently, alternative processes have been discussed in the literature to produce (bio)hydrogen—hydrogen produced from biowaste and using fermentation processes [23]. A comprehensive discussion comparing the production processes, and economics of hydrogen produced by electrolysis, biowaste refining, and algae biodiesel indicates that hydrogen produced from biowaste has a lower total cost than the other two [132]. On a side note, we identified several works [42,43,64,98,101] that propose optimization models that rely on waste to produce hydrogen—these works are discussed in the remaining text according to the features of their models.

Today hydrogen is primarily produced via SMR given that the technology is well established in the market and the production process is relatively inexpensive [3,10,133]. Nevertheless, SMR results in significant greenhouse gas emissions, namely carbon dioxide and methane (leaking).

Electrolysis is expected to grow and eventually overtake SMR-based technology as it offers a low-carbon means to produce hydrogen. Electrolysis will have a low carbon footprint if its electricity requirement is met by renewables [3]. Low-temperature electrolyzers, such as alkaline, and proton exchange-membrane (PEM) are commercially available and high-temperature solid-oxide electrolyzers are at an advanced stage of demonstration [134]. Alkaline and PEM electrolyzers have lower capital costs than solid-oxide electrolyzers, however, electrolyzer technology is still significantly more expensive than fossil fuel-based production. Investment costs have been decreasing and are expected to continue to drop as the costs of renewable technologies also go down [3].

Alkaline electrolyzers are an established reliable technology characterized by slow start-up capabilities [135]. PEM electrolyzers have become commercially available in recent years. They produce the highest purity of hydrogen gas [136]; however, they require rare metals such as platinum and iridium as well as complex membrane structures to operate [135]. Given the rarity of iridium, it may be challenging to scale up PEM capacity globally. Solid oxide electrolyzers have the highest efficiencies but a long start-up time [137]. They also require high operating temperatures and thermal cycling which may result in material degradation [137]. Ursua et al. [138] suggested that electrolytic systems especially alkaline electrolyzers that operate on renewables should be coupled with battery systems to capture excess power and thus maintain hydrogen production during periods of low renewable output [138].

Gasification is an efficient process that has been utilized for hundreds of years—as early as the 1800s [3]. It can use biomass or coal as a feedstock and does not require complex technology. Biomass gasification is attractive because biomass is a renewable resource that recycles carbon dioxide and results in low net greenhouse gas emissions. However, a significant limitation of biomass gasification is the availability of feedstocks, which is mainly regionally dependent. An alternative to biomass is coal, which is more abundant and relatively cheaper compared to biomass and even natural gas. However, a deterrent to expanding coal gasification for hydrogen production is the release of greenhouse gas emissions.

Lesser used technologies and processes found in the literature include thermochemical water splitting, the dry reformation of methane (DRM), and pyrolysis. Like electrolysis, thermochemical processes use some energy source to split water to produce hydrogen (and oxygen as a byproduct). Thermochemical water splitting relies on high-temperature thermal energy (from concentrated solar or nuclear) to drive a series of chemical reactions, such as the sulfur-iodine (S–I) cycle and copper-chlorine (Cu–Cl) cycle, and consequently produce hydrogen with near-zero greenhouse gas emissions. However, the limiting factors for thermochemical processes today include the need for efficient and robust reactor designs, the high costs of concentrated solar systems and the availability and costs of critical materials (for specific thermochemical cycles) such chromium, copper, nickel, and zinc. In the literature, one paper [108] models hydrogen production via a solar thermochemical system, and one paper [55] explores hydrogen production by a nuclear-based S–I thermochemical system. Deng et al. [139] performed multi-attribute decision analysis methods, to rank different thermochemical cycles. They found that $MnO/Mn_3O_4/Na_2CO_3$ and $MnFe_2O_4/Na(Mn_{1/3}Fe_{2/3})O_2$ cycles ranked the highest due to the moderate temperature reaction requirements, the low cost of reactants, and the ease to prepare said reactants. A later study by Liu et al. [140] performed experiments of the $MnO/Mn_3O_4/Na_2CO_3$ cycles in an electric furnace and concentrated solar furnace to access the economics of the process. The authors found that 1) the concentrated solar furnace resulted in a conversion efficiency of 95% and 2) to achieve competitive hydrogen prices of $2/kg over 80 chemical cycles should be used.

In the literature, the use of DRM [37] was coupled with. Bacteria use anaerobic digestion to break down biomass to create methane-rich biogas (with CO_2) which is then reformed to produce hydrogen. The process uses carbon dioxide as the oxidizing agent for methane to produce hydrogen. However, the process is highly endothermic, given the stability of CO_2. Furthermore, biomass as a feedstock is limiting for the reason described for biomass gasification.

Pyrolysis technology has been widely established and commercially available for years. For hydrogen production, the process involves the thermal decomposition of methane. The primary benefit of pyrolytic processes is that there are no carbon dioxide emissions [141]; the output is hydrogen gas and solid carbon; One paper [52] we surveyed, modeled hydrogen production from methane pyrolysis. Nevertheless, there has been a push in Europe, particularly Germany, to develop efficient ways to produce hydrogen via pyrolysis.

Byproduct production of hydrogen is common in the refining, chemical, iron, and steel industries. In refineries, catalytic reformers convert petroleum naphthas into high-octane liquids products [44,68,84,105]. In the process, hydrogen is produced as a byproduct that a refinery can use for hydrocracking purposes or to reduce the sulfur content of diesel. In the chemical sector, Chlor-alkali processes that produce chlorine and sodium hydroxides also produce byproduct hydrogen [61,116,122]. Steel production is another industry mentioned in the literature that produces byproduct hydrogen [122]. In the production of steel, coal is carbonized to produce coke; the process also produces byproduct coke oven gas composed of hydrogen, methane and nitrogen, carbon dioxide, and carbon monoxide. The hydrogen in the gas can be extracted and injected into the hydrogen supply chain. Yoon et al. [125], and Stiller et al. [112] optimized a hydrogen supply chain utilizing byproduct hydrogen in addition to hydrogen produced by more conventional production methods. Nevertheless, the authors did not specify the source of the byproduct hydrogen.

A summary of the hydrogen production technologies explored in the literature is shown in Fig. 4; a more detailed classification is given in Supplementary Datasheet 1.

Carbon capture, transport, utilization and storage

Carbon emitting hydrogen production plants such as SMR, and coal and biomass gasification, can be retrofitted with carbon capture technologies to reduce the amount of greenhouse emissions [142]. Captured emission can be stored in geological formations such as natural saline aquifers and depleted oil and gas reservoirs, or in the deep ocean [143]. If not stored captured carbon dioxide can be recycled and used

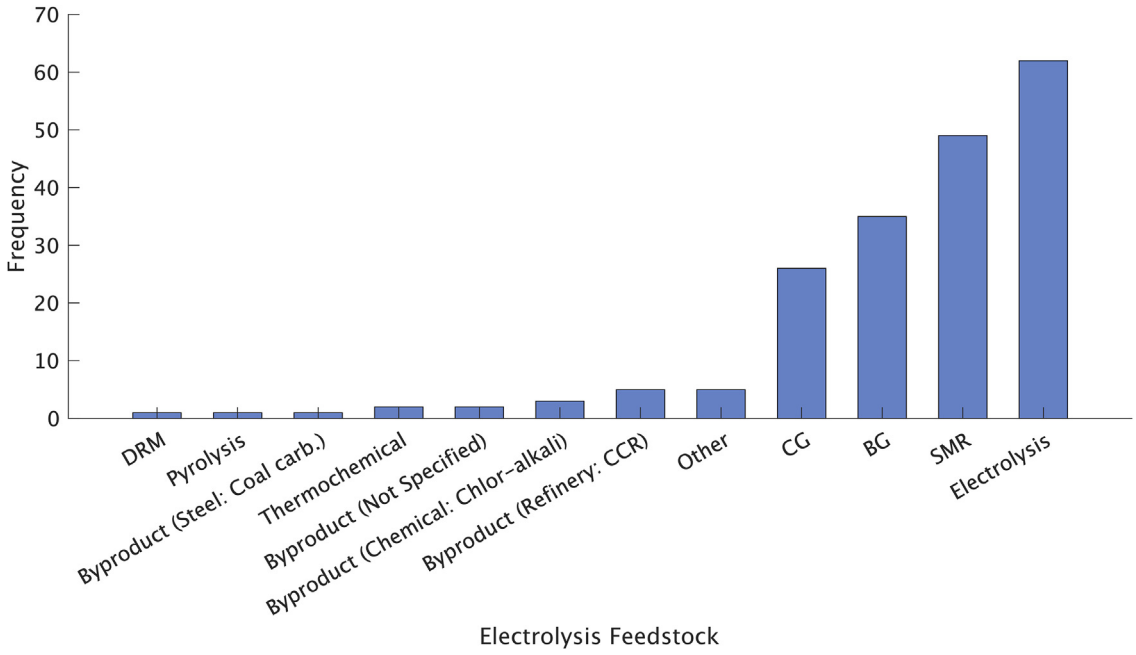


Fig. 4 – Technology used for hydrogen production in the literature. Electrolysis, steam methane reformation (SMR), coal gasification (CG), and biomass gasification (BG) were the most common hydrogen production methods found. Byproduct hydrogen was sourced from refineries via continuous catalytic reforming (CCR), the chemical sector via chlor-alkali processes, and the steel sector via coal carbonization for coke formation. Less common methods include dry reformation of methane (DRM), pyrolysis, and thermochemical water splitting.

in enhanced oil recovery or the production of carbon-based e-fuels [144]. Carbon capture will require a transportation network to move emissions from sources to sinks, either storage or other users. Therefore, it is necessary to understand the configuration of such a transportation network and properly assess the cost-effectiveness of carbon capture technologies. In the literature we reviewed, Agnolucci et al. [28] allow their model to build carbon dioxide pipelines that transport emissions from production nodes in the UK to storage sites. Almansoori and Betancourt-Torcat [29] do the same for their case study in Germany.

Many of the models found in the literature with carbon intensive hydrogen production allow for investments in new plants that have carbon capture and storage capabilities. However, it will be necessary to allow models to retrofit existing plants as well. This will give a more realistic representation of real-world options.

Furthermore, the application of CCS technology will require additional investment and operating costs to retrofit plants or build new state of the art production facilities. It is important to note that production plants with CSS technology will therefore have higher hydrogen production costs compared to plants without. Table 2 shows the LCOH of SMR, CG and BG plants with and without CCS.

Transportation

Hydrogen can be transported as a liquid, a compressed gas, or using a hydrogen carrier, namely ammonia or some liquid organic hydrogen carrier (LOHC). Liquefied hydrogen can be transported on tanker trucks, tank railways, or ships, whereas gaseous hydrogen may be transported via high-pressure pipelines, tube trailers, or tube railways. The transportation of hydrogen can be divided into transmission and distribution. Transmission occurs across long distances between production sites and storage terminals, often within regions, across a country, or several countries. Transmission networks are made up of all previously mentioned transport methods. On the other hand, distribution networks are made up of pipelines or trucks that transport hydrogen to a consumer. Hydrogen distribution is often neglected in the HSCD problem.

The transportation mechanism used is highly dependent on the investment and operating costs and the demand distribution, which determines how far hydrogen needs to be transported. Long-distance delivery may require pipelines or shipping vessels. However, a pipeline may be infeasible or too costly to use for the transport of hydrogen across the ocean. Another factor to consider concerns hydrogen losses. Liquid hydrogen transport is prone to appreciable losses as it must be cryogenically stored, and therefore it is susceptible to a significant amount of boil-off.

Fig. 5 depicts the number of times a transportation method was modeled in the papers surveyed. Transportation via trucks (liquid and gas) and pipelines was most commonly modeled in the literature, while only five papers included shipping vessels as a transport medium. However, as countries develop long-term plans to export hydrogen to other nations, it is necessary to consider modeling the international trade of liquid hydrogen or other hydrogen carriers.

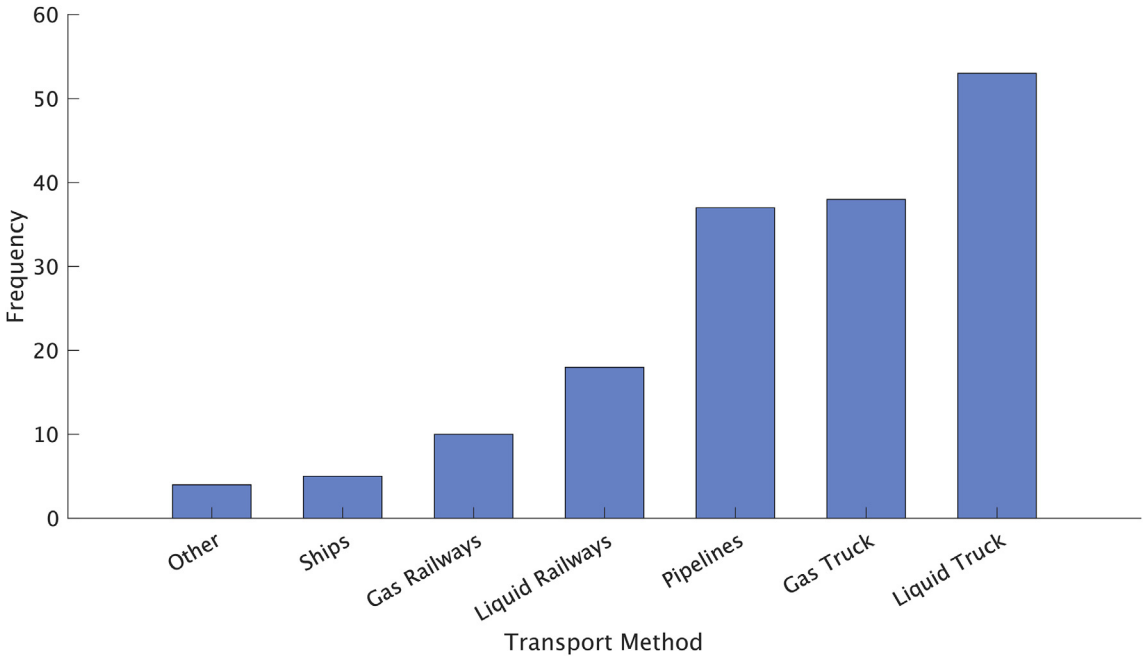


Fig. 5 – Hydrogen methods modeled in the literature. Trucking (gas and liquid) and pipelines were the most common hydrogen transport methods.

Regarding pipeline transportation, dedicated hydrogen pipelines are currently used but the networks are relatively small [6]. Currently, the United States has about 2600 km of hydrogen pipelines [145] compared to the 300 thousand km of transmission lines that make up its natural gas network. One topic of discussion, although it is not explored in the literature we reviewed, is the idea of using natural gas pipelines for hydrogen transport or blending hydrogen into natural gas. These retrofits or blends offer a cheaper alternative to building an entirely new transportation network [6]; however, they do come with their own set of challenges. Existing natural gas pipelines would require retrofits; blends with hydrogen less than 15% would require only minor modifications [146]. Shares greater than 40% will require compressors to be replaced while pure hydrogen transport would require lines to have more (and new) turbines and compressors to deliver the hydrogen an appropriate volumetric flow given its volumetric energy density [147]. Another problem with using natural gas pipelines for hydrogen transport is that existing infrastructure is more than a century old and leak prone [148].

Shipping vessels are generally utilized for international trade, which is not commonly explored in the literature. Currently, long distance international trade is not an economical option given the low volumetric energy density of hydrogen and high production cost when using renewables; however long-distance ship transport as liquid hydrogen or as liquid organic hydrogen carriers (LOHC) may be the least costly mechanism [6]. Only two papers took an international perspective, allowing hydrogen to be transported via shipping vessels. Heuser et al. [7] looked at hydrogen transport between Argentinean Patagonia and Japan. The authors considered transporting hydrogen from production sites to a harbor storage terminal via pipelines within Argentina. They determined the number of liquid hydrogen shipping vessels needed to transport hydrogen between the countries in a year, considering boil-off, the loading and discharge times of hydrogen into the ships, and the duration of the trip.

Hong et al. [62] performed a techno-economic analysis of the hydrogen supply chain within the ASEAN region, from Malaysia, Indonesia, and Australia to Singapore. They explored hydrogen transmission as a gas, liquid, ammonia, or methylcyclohexane (MCH). Because of the geography of the region, shipping is a common transport medium, and several shipping-based supply chains were explored in the paper.

Shipping vessels were also explored for national hydrogen transport for countries with islands or offshore regions. Sabio et al. [102] and Sabio et al. [103] allowed for investments in shipping vessels between mainland Spain and the Balearic Islands, the Canary Islands, Melilla, and Ceuta. They noted that these regions had to import hydrogen via freighted ships or remain isolated and produce all hydrogen locally.

End-users/demands

End-users are entities that will utilize hydrogen in some capacity. Currently, most hydrogen is consumed by refineries or ammonia facilities [10]. Four papers [65,68,84,105] explore the design of stand-alone hydrogen networks to satisfy refinery demand. These papers focused on localized/onsite hydrogen production, typically via CCR or SMR.

Most of the literature reviewed for supply chain design problems model the hydrogen network for the transportation sector, specifically fueling stations, with demand projections often dependent on the market penetration of HFCVs. Even in cases where fueling stations are not modeled explicitly, hydrogen demand estimates remain largely based on projections from the transportation sector.

Table 3 – Papers with hydrogen demand projections from more than one sector.

Reference	Transport	E-fuel	Refinery	Power	Commercial	Industrial	Residential
Carrera & Azzaro-Pantel (2021)	•	•					
Carrera & Azzaro-Pantel (2021)	•	•					
Contaldi et al. (2008)	•		•	•			•
Dagdougui et al. (2012)	•					•	
Demirihan et al. (2020)		•		•			
Demirihan et al. (2021)		•	•	•			
Heuser et al. (2019)	•		•	•			
Hong et al. (2021)	•			•		•	
Kazi et al. (2021)	•					•	
Mah et al. (2020)	•			•			
Mah et al. (2022)	•			•			
Ogumerem et al. (2019)	•	•		•			
Qadrdan et al. (2008)	•			•			
Shamsi et al. (2019)				•		•	
Yáñez et al. (2018)	•				•		•

Other papers explore the role of hydrogen in power-to-power systems, i.e., using hydrogen as an energy storage medium for the power sector. Chen et al. [41] model a wind-hydrogen-fuel cell microgrid that utilizes hydrogen to ensure optimal power dispatch. In this case, hydrogen demand is only from the power sector. Jiang et al. [69] integrate a hydrogen supply chain and a generation and transmission planning power system model to produce transportable hydrogen that can be stored to ease seasonal electricity imbalances.

Most papers consider hydrogen demand from a single sector, however, the push to utilize hydrogen as means to decarbonize various sectors [149,150] leaves many hydrogen demand projections in the literature largely underestimated. Taking into consideration hydrogen demand from multiple sectors such as heating (via cogeneration using hydrogen as a feedstock), the electric power system (hydrogen as an energy storage medium or for power generation), and chemical and industrial (steel, paper, cement), etc., will provide a more accurate representation of how a hydrogen economy would develop [151]. However, only a few of papers take such an approach as depicted in Table 3.

For example, Demirhan et al. [51,52] consider the role that hydrogen will play in the power sector as a power generating and energy storage medium. They also explore the use of hydrogen to produce chemicals and thus formulate their model to include hydrogen production for refinery use and the production of alternative fuels. Carrera and Azzaro-Pantel [38,39] consider hydrogen as means to produce methane in addition to demand by HFCVs. Contaldi et al. [44] use the MARKAL-Italy model to find the optimal hydrogen supply chain configuration to meet hydrogen demand for transportation, refinery use, the power sector via combined cycle hydrogen turbines, and residential/commercial heating. In the same respect, Yañez et al. [122] include hydrogen demand from the commercial, residential, and transportation sectors.

Modeling features

Decisions

Hydrogen supply chain design often involves investment and operational decisions to ensure demand is adequately met. Investment decisions include a) the types, locations, and capacities of facilities (hydrogen production facilities, hydrogen storage sites, carbon capture technology, power generation plants) to build; and b) the transportation network. These investments may be allocated to alternative periods, enabling the investment to occur, for example, in different years. Operational decisions include the amount of hydrogen that facilities will produce at a given time, which end-users the hydrogen will be sent to, and which transportation route. These decisions depend on input parameters such as resource availability and hydrogen demand. Operational decisions are used to provide feedback to investment decisions, to ensure that an optimal network configuration is obtained. The resolution of operational decisions can play a significant role in determining the production capacity of a system. Further discussion on the operational resolution is found in the section on Temporality below.

Performance indices

Financial

The most common performance measure optimized is the total cost of the hydrogen supply chain. At the core of the cost is the capital investment cost, the operating cost of production, storage, and transportation technologies. In efforts to quantify the cost of emissions, some authors have sought to incorporate the cost of CCS technologies and the cost of building and operating a CO_2 pipeline network. The cost of carbon has been directly measured by expanding the cost definition to include carbon tax costs as well as the cost of emission trading schemes. Han et al. [60] explore the mitigation cost of implementing carbon emission trading schemes and installing carbon capture and storage within the HSC. In such a model, the decisions favor installing non-fossil-based hydrogen production or investing in a mitigation strategy that drives the mitigation costs down.

Additional cost measures include the cost of feedstocks and resources such as real estate and land costs for building new facilities [64,75,78,121]. Hwangbo et al. [64] introduce a scaling factor that considers real estate fees to account for variations in each region. The real estate fees are calculated with a cumulative distribution function based on the mean and variance of the costs.

Kim & Kim [75] directly considered land costs in the objective function. The capital costs of land are determined by multiplying the area occupied by a production or storage facility, the number of facilities built, and the regional place.

Other papers [74,96,98,101] seek to maximize profit as their performance measure. Ogumerem et al. [96] formulate their problem to maximize the profit, defined as the difference between the revenues obtained from selling hydrogen and by-products and the total operating and investment costs. Reyes-Barquet et al. [101] maximize profit, a function of total hydrogen sales at storage and dispatch stations minus total supply chain costs.

Financial risk is an important performance measure as it considers the possibility of low profits or high costs. For example, Sabio et al. [102] took a multi-objective programming approach to assess the trade-offs between the risk of high costs and the design and operating costs of a hydrogen supply chain. Hwangbo et al. [64] proposed a two-stage stochastic model that considers financial risk due to fluctuations in the expenses resulting from uncertain demand conditions. They calculate an index for downside risk, which collects costs above an expected or target value.

Environmental

Assessing the environmental impacts of a potential HSC involves addressing which environmental factors are necessary to consider in a modeling strategy. Given the growing concern of rising greenhouse gas emissions, total emissions along the supply chain or per unit emissions are the most common measures used [63,92,96]. Global warming potential (GWP) is another indicator applied in the literature [48,50,93,94,101] that offers insights into the overall effect of the heat radiation absorption of the atmosphere due to greenhouse gas emissions. Often the GWP of different stages of the HSC is calculated and the total system GWP is minimized.

The literature we reviewed generally assesses the GWP of traditional greenhouse gas emissions such as CO2 and methane. Recently there is a growing concern of the role hydrogen can play as a greenhouse gas by causing indirect warming effect. A report by the UK Department of Business, Energy and Industrial Strategy has shown that hydrogen released/leaked into the atmosphere would cause a decrease in hydroxyl radicals that would increase the atmospheric lifetime of methane (GWP of 2–30 over 100 year) [152]. Hydrogen leakage will also significantly increase water vapor in the atmosphere. Fan et al. [148] assert that a focus should be placed on leakage in a developing hydrogen economy and address those issues from the start. It is necessary to quantify the potential leak rate throughout the hydrogen supply chain so to assess its GWP. Therefore, a relevant environmental performance metric that should be considered is the minimization of the leak rate of the hydrogen supply chain. The current GWP metrics should also consider the indirect GWP of hydrogen.

Other environmental factors include life cycle indicators. Guillen-Gosalbez et al. [57] consider damages to human health by climate change measured in disability-adjusted life years. Noting that life cycle assessment criteria can be used to consider environmental factors, Sabio et al. [103] extend the work of Guillen-Gosalbez et al. [57] by adding ecological factors including damages to mineral and fossil fuel resources and ecosystem damage caused by acidification, eutrophication, and ecotoxic substances.

Ochoa-Robles et al. [94] propose a social cost-benefit framework to assess the external cost that has to be included from several environmental factors. In addition to greenhouse gas emissions, they considered platinum depletion (related to the manufacturing of hydrogen fuel cells), air pollutants (such as NOx, carbon monoxide, and hydrocarbon), and noise pollution.

Safety

Hydrogen momentum declined for some time due to safety concerns. These concerns resulted from incidents under specific conditions that resulted in leakage and explosions of fueling stations. Therefore, safety measures have been deemed necessary in HSCD. A few papers have discussed safety risk assessment approaches for hydrogen. Kim and Moon [77] proposed a framework for the HSCD risk analysis, which consists of the following steps: hazard identification, consequence and frequency analyses, risk evaluation, and the development of a relative risk index.

Additional performance measures

Some hydrogen supply chain problems address the location setting of refueling stations. While some papers focus on the cost minimization of the system, others optimize performance measures that ensure maximal coverage by the stations. Bae et al. [36] sought to minimize the distance fuel cell vehicles travel to a station to ensure the optimal location for refueling sites. In the same respect, Li et al. [80] maximized the total number of vehicles serviced by refueling stations.

Other performance measures explored in the literature on HSCD problems include maximizing the total renewable methane produced with the hydrogen supply chain. Carrera & Azzaro-Pantel [39] introduced a renewable methane constraint to reduce natural gas imports by increasing the share of synthetic methane, with hydrogen from renewable sources being used to meet methane demand.

Multi-objective

Multi-objective optimization involves making decisions considering trade-offs between two or more conflicting objectives. Most models in the literature take a single-objective approach, most often optimizing a financial measure such as cost. However, doing so fails to neglect other important objectives to consider in developing a hydrogen supply chain. Taking only cost minimization into account may result in a system that is not environmentally friendly – opting to invest in the cheapest technologies and avoiding cleaner, more expensive alternatives such as electrolysis and SMR with CCS. The converse is also true; the most eco-friendly system may not be the cheapest.

Reyes-Barquet et al. [101] explored the configuration of an agro-industrial waste-powered hydrogen supply chain in Mexico that considered the need to maximize profit and minimize the GWP. The authors first solved their models with a single optimization approach using both objective functions. They found that under a GWP minimizing scheme, GWP decreased by 20%, but the net profit margin also decreased by 20%.

Twenty-three of the papers we reviewed considered multi-objective optimization. Table 4 summarizes the papers that

Table 4 – Classification of papers that take a multi-objective approach.

Reference	Financial			Environmental			Safety	Other[a]
	Cost	Profit	Risk	GWP	Emissions	LCI	Hazards	
Carrera & Azzaro-Pantel (2021)	•			•				
Carrera & Azzaro-Pantel (2021)	•				•			•[a]
Dagdougui et al. (2012)								•[b]
DeLeon-Almaraz et al. (2013)	•			•			•	
DeLeon-Almaraz et al. (2014)	•			•			•	
DeLeon-Almaraz et al. (2015)	•			•			•	
Fazli-Khalaf et al. (2020)	•				•			•[c]
Guillen-Gosalbez et al. (2010)	•					•		
Han et al. (2013)	•						•	
Hugo et al. (2005)	•				•			
Hwangbo et al. (2018)	•		•					
Kim & Moon (2008)	•						•	
Li et al. (2008)	•				•			
Lim et al. (2020)	•				•			
Ochoa Bique et al. (2019)	•				•		•	
Ochoa Robles et al. (2019)	•			•				
Ochoa Robles et al. (2020)	•			•				
Ogumerem et al. (2018)		•			•			
Ogumerem et al. (2019)		•			•			
Reyes-Barquet et al. (2022)		•		•				
Sabio et al. (2010)	•		•					
Sabio et al. (2012)	•					•		
Weber & Papageorgiou (2018)	•						•	

GWP: Global warming potential; LCI: Life cycle indicator.
[a] Renewable methane produced.
[b] Maximum stored hydrogen, demand satisfaction, and power sold to the grid.
[c] Supply chain reliability, and social responsibly.

take multi-objective approaches with a breakdown of the performance indices that are considered.

Uncertainty

Deterministic models inherently assume perfect information regarding the input parameters. Such an approach, however, fails to consider the uncertainties arising in a real-world setting, which may render the solution to the deterministic model infeasible when input parameters vary.

Sources

Within the context of the hydrogen supply chain design problem, there are four general sources of uncertainty: demand, technological specification, cost/price, and feedstock-related uncertainties. Uncertainty in hydrogen demand was most commonly modeled within the literature; see Table 5 for a summary of the sources of uncertainty found throughout the literature reviewed. General hydrogen demand projections have typically been extrapolated using assumptions

Table 5 – Sources of uncertainty found throughout the literature reviewed.

Source of Uncertainty	Parameter	References
Demand	Transportation	[32,47,78,90,93,95,124]
	Refinery	[66,68]
	Not specified	[53,64,77]
Technological Specification	Facility capacity	[53]
	Purifier capacity	[84]
	Per pass conversion	[68]
	Pipeline flow rates	[84]
	Efficiency	
	Current/Voltage	[45]
	Membrane thickness/area	[45]
	Operating temperature	[45]
Financial	Capital cost	[45,53]
	Operating cost	[45,53,102]
	Feedstock/raw material costs	[65]
	Hydrogen price	[68]
	Interest rate	[45]
Feedstock	Availability/Variability	[45,87,124]

about the market penetration of hydrogen fuel cell electric vehicles [32,47,78,90,93,95,124]. This may represent a good baseline for hydrogen demand; however, demand may be underestimated as hydrogen is expected to penetrate other sectors besides transportation. Using realistic assumptions of potential hydrogen usage in heating, power, and chemical sectors, as well as transportation, may provide better estimates of future hydrogen demand.

For hydrogen-refinery-specific networks, demand uncertainty depends on the amount of hydrogen necessary to increase the purity of fuels (sulfur removal). Jagannath et al. [68] consider demand uncertainty at the inlet of a processing unit. The demand was directly dependent on the purity of the crude oil. Hwangbo et al. [66] also consider the uncertainty of hydrogen demand in a petrochemical complex. However, the authors also extend their model to consider uncertainty in electricity, steam, and water demand.

Uncertainties in technological specifications are also necessary to consider as they impact the production output of hydrogen. The conversion efficiency, the amount of feedstock needed to produce one unit of hydrogen, can change in the future due to advances in research and development. None of the papers we reviewed explored the uncertainty of future efficiency projections of hydrogen production technologies. Uncertainty in process operations such as reaction per pass conversion [68], pipeline flow rates [84], and purifier capacities was considered in the refinery context. The capacity for hydrogen production technologies was considered by one study [53].

Technology costs generally dictate the hydrogen production mix determined by a planning model. Introducing uncertainty in the costs generates solutions that consider probabilistic cost outcomes that may influence the solution of the model. For instance, the capital cost of electrolyzers is expected to decrease, as well as the cost of renewable technologies, particularly wind and solar. A model that fails to capture the probability that the capital cost of electrolyzers may decrease in the future may underestimate how much electrolyzer capacity should be installed compared to other technologies. Among the papers surveyed, only Fazli-Khalaf et al. [53] consider the capital cost of hydrogen as a source of uncertainty in their model.

Operating costs and feedstock costs are additional cost uncertainties that can be considered. Two papers explored uncertainty in operating cost [53,102]. Hwangbo et al. [65] explore the feed cost/raw material cost inputs; they considered the material cost of fuels, industrial water, and electricity. Coppitters et al. [45] consider 18 uncertain parameters for the HPD model of a PEM electrolyzer system. They consider uncertainty in electrolyzers technical parameters such as the membrane thickness and area, and the limiting current. For the PV cells, the technical specifications considered were the open circuit voltage and short-circuit current. The authors also consider uncertainty in the ambient temperature and solar irradiance, both of which determine the power output of the PV cell. Finally, the economic parameters include the capital and operating cost of both the electrolyzer and PV units as well as the discount rate.

Modeling approaches

Most studies considering uncertainty take a stochastic programming (SP) approach with a two-stage decision framework; see Table 6 for a list of the modeling approaches found in the literature reviewed. In such models, scenarios of uncertain parameters are constructed by discretizing their probability density functions [153]. In a SP model, the two-stage decision framework considers first-stage decisions taken prior to knowing which scenario will be realized and second-stage decisions that are scenario-dependent. The advantage of SP is that the solution obtained is feasible under the realization of any of the scenarios in the model. The disadvantages of SP are that the probability distribution of these uncertain parameters is not always known, and the models may become intractable.

Table 6 – Modeling approaches for problems dealing with uncertainty found in the literature reviewed.

Stochastic Programming	Fuzzy Programming	Chance Constraint Optimization	Robust Optimization
32, 47, 64, 65, 66, 68, 77, 78, 84, 87, 90, 102, 124	53, 93	124	45, 84

In fuzzy programming, uncertain parameters are treated as fuzzy numbers. Fazli-Khalaf et al. [53] take a fuzzing programming approach in which they had four fuzzy numbers: investment cost of new plants, variable operating expenses, hydrogen demand of customer zones, and the maximum facility processing capacity. Ochoa Robles et al. [93] also model hydrogen demand uncertainty using fuzzy concepts. The authors state that in the context of hydrogen supply chains, fuzzy programming is superior to stochastic programming because the latter requires historical data to determine different hydrogen demand scenarios; to date, historical demand for hydrogen is not available for most sectors (excluding refineries and ammonia facilities).

Chance constraint programming (CCP) [153] enables the definition of constraints satisfied within a given probability threshold. Yang et al. [124] use CCP to deal with the uncertainty of wind power to reduce curtailment by wind farms in off-grid HSC in Fujian Province, China.

Lou et al. [84] use a robust optimization (RO) framework to optimize the hydrogen network of refineries given the uncertainty in the hydrogen source and sink flow rates of six utilities. The RO approach developed by Mulvey and Zenios [154] integrates goal programming formulations with the scenario-based description of the problem to generate robust solutions. They produce a series of solutions that are progressively less sensitive to realizations of the model data from a scenario set. The authors compare the RO formulation of their problem to an SP formulation.

Temporality

Investment strategy

Hydrogen supply chain network design problems involve determining the system and infrastructure layout necessary to meet the demand for a given time horizon. There are two approaches: 1) single-stage investment and 2) multi-stage investment models. With single-stage investment models, a

time horizon of multiple years is represented by one target year, and the model makes investment decisions regarding technologies for that year. The target year is the first and last year of the time horizon. A single-stage investment model, or a snapshot approach as Almansoori and Shah [17] label it, represents the system configuration at the end of the time horizon.

The clear advantage of using a single-stage investment approach is the model size reduction. However, using a target year prevents multiple-stage investments during the time horizon that could be used to account for variable input parameters during the time horizon, for example, increasing demand or changes in technology costs.

On the other hand, multi-stage investment models break down the entire planning horizon into several investment stages (e.g., annual, biannual, every five years), where investment decisions are made at the start of each stage until the end of the time horizon. Multi-stage models have the benefit of providing when various technologies along the time horizon would be built, namely by capturing the hydrogen demand growth trajectory. Nevertheless, using multiple stages requires additional variables and constraints that may render the model intractable.

HPD problems were modeled using a single-stage investment approach. However, HSCD problems were split between single-stage (in 44 papers) and multi-stage (in 38 papers) investment models.

Operational resolution

Hydrogen supply chain problems incorporate some representation of operational decisions where facilities produce hydrogen to meet demand in a given period. In the literature, most papers use a per-day average demand of hydrogen for a given year or the total annual demand. While this is common practice, such an approach neglects to consider interannual demand variability and feedstock availability. Variability becomes more critical in systems that will produce hydrogen from renewable resources, many of which are intermittent and vary throughout the day and between seasons of the year. For such systems, finer resolutions than average yearly operations should be considered to ensure a reliable infrastructure configuration.

Papers that explore the role of biomass as a feedstock for hydrogen production generally have a more refined operational resolution, namely monthly [42,59,119,121]. Biomass availability varies seasonally and according to the biomass type. Won et al. [119] proposed a model for a renewable energy source-based hydrogen system considering six different biomass types and wind and solar as feedstocks for hydrogen production. The authors note that biomass availability from urban and forest residue is relatively constant throughout the year. However, biomass from rice straw and husk is abundant between August and November, and barley straw biomass is abundant between March and June. Noting the seasonality of various biomass feedstock, the authors used a monthly time resolution to capture biomass availability throughout the year.

Renewable resources such as wind and solar have seasonal and daily variations. Therefore, finer resolutions are necessary to capture said variability adequately. Yang et al. [124] sought to capture wind power and hydrogen demand variability by using an hourly resolution for one day taken to represent an entire year. Using Monte Carlo simulations, they generate a sample set of daily hydrogen demand (with an hourly resolution). The authors then apply density-based spatial clustering of applications with noise on the demand samples to obtain three representative profiles and their respective probabilities for stochastic programming.

Demirhan et al. [51] model hydrogen production for renewable power generation and energy storage. The authors state that successfully integrating renewables into the power sector requires capturing the daily and seasonal variability of intermittent resources. Noting that modeling all 8760 h of a year can render the model intractable, they use agglomerative hierarchical clustering on the renewables, electricity prices, and power load to obtain a set of days with hourly resolutions to represent an entire year. They subsequently run the model using various numbers of representative days to get the optimal infrastructure configuration and operation of their network. Demirhan et al. [52] extend the previous model and apply the same clustering approach for a system that uses hydrogen as a feedstock for power generation and energy storage, chemical production (e.g., methanol and ammonia), and subsequently fuel (e.g., methane, diesel, gasoline) production.

Solution methods

The HSCD and HPD problems have distinct features regarding decisions, time scales, uncertainty, and mathematical properties. HSCD problems involve larger time horizons, coarse time resolutions, and uncertainty. The problems are usually linear and involve continuous and binary or integer variables. These problems are most commonly formulated as MILP problems, see Table 1. On the other hand, HPD problems have shorter time horizons, a finer time resolution, and no uncertainty and often involve nonlinear constraints, resulting in NLPs or MINLPs and simulation-based problems. Therefore, the solution methods applied are inherently distinct.

HSCD and HPD can be solved either directly or using decomposition methods that exploit a spatial dimension, time dimension, or feature (e.g., scenarios, facility type) that can decompose the problem into a master problem and subproblem(s). There are general decomposition approaches for specific problems, such as the L-Shaped algorithm [155] or Benders Decomposition [156].

In addition to decomposition methods, rolling horizon approaches decompose the original problem, based on periods, into subproblems, which are mainly suitable for problems integrating multiple time scales (e.g., demand growth over the years and an hourly resolution for renewable sources). These approaches are used to cope with the complexity of large-scale problems by solving smaller problems, some at the cost of finding sub-optimal solutions.

Hydrogen supply chain design

Single-objective problems are typically solved using a specific algorithm or a general solver. These can be further classified in terms of the solution, which can be optimal or a heuristic. Using a heuristic helps find approximate solutions without exhaustively finding every solution. The solution can be the

optimal solution (or close to the optimal solution) or a heuristic solution.

In the papers reviewed, single objective deterministic HSCD problems were generally MILP problems or MILP reformulations that were solved using a commercial solver that typically implements a branch and bound algorithm. Branch and bound involves partitioning a set of feasible solutions into smaller subsets and systematically evaluating them until the best solution is found [157,158]. Six papers [44,55,69,100,113,114] solved an LP model; the commercial solver implements a Simplex method algorithm or some variant.

Single-objective problems can also involve uncertainty. In the literature, all single objective stochastic HSCD problems were formulated as MILPs and solved directly using a commercial solver. However, adding stochasticity may result in a large model that may require decomposition methods to be solved more efficiently (shorter computation times).

A summary of the solutions methods for HSCD problems (deterministic vs. stochastic; multi-objective) is provided in Supplementary Datasheet 1.

Hydrogen process design

HPD optimization problems are typically nonlinear that can be solved using derivative-based methods, derivative-free algorithms (e.g., metaheuristics), simulations, or decompositions methods. Commercial solvers have also been developed to solve complex nonlinear optimization problems.

Refinery hydrogen network problems are HPD problems that determine the optimal production and allocation of hydrogen resources to satisfy the demand for refinery processes. These models are either deterministic [68,105] or stochastic [84]. Lou et al. [84] formulated their model as an MINLP and solved using the DICOPT [159] solver.

Sardashti et al. [105] formulated their problem as a deterministic NLP and an MINLP and solved using two different global optimization algorithms involving McCormick [160] envelopes. The MINLP is solved using a bound contraction procedure proposed by Faria and Bagajwicz [161]. The framework of this approach is based on updating bounds for each variable one at a time to find a globally optimal solution.

Jagannath and Almansoori [68] explored the optimal configuration of a hydrogen refinery network considering uncertainty in its operating conditions. The authors formulated their model as an MINLP, which they solved using a global optimization algorithm, similar to the one proposed by Teles et al. [162] and Castro and Teles [163]. They compared their approach to solving the model with BARON [164] and DICOPT [159] to solve the MINLP directly. Their solution approach resulted in significant computational time savings compared to the direct solvers.

Hwangbo et al. [65] designed an integrated utility (petrochemical complex) and hydrogen supply network formulated as an MINLP. They solved their model using an iterative relaxation scheme. For the solution approach, the authors initially assume that the capital and operating costs of SMR process plants are not affected by location. They determine the coordinates of candidate SMR locations to calculate regional total costs based on the distance between the SMR plant and the utilities. Nonlinear terms are replaced with artificial linear terms, which determine where to build pipelines and the flow rate within the pipes. A modified total cost objective function is minimized until the smallest modified cost of the system is obtained.

Khalilnejad et al. [73] design a photovoltaic-alkaline electrolytic system formulated as a multilevel optimization problem solved using a genetic algorithm. In the first level, they optimize the dimensions of the system (i.e., the size of the electrolyzer stacks, and photovoltaic capacity), while in the second they use the optimal temperature of the system to optimize hydrogen production and power losses (excess PV power production). Similarly, Coppitters et al. [45] solve their HPD model of a photovoltaic-PEM electrolytic system using a genetic algorithm.

Saur and Ramsden [106] use the H2A model by NREL to optimize a wind-based electrolytic system. H2A is a spreadsheet model that performs technoeconomic calculations of various wind-electrolyzer configurations given various electricity pricing schemes. The model determines whether to use wind power to produce hydrogen or sell to the grid.

Other HPDs were solved using simulations [107] and techno-economic analyses [107].

The HPD models that use MILP formulations [59,74,109] are solved using a commercial solver, or the solution method is not mentioned. Chen et al. [41] are the exception; they use particle swarm optimization and a genetic algorithm to solve their HPD problem.

An important paper to note is Hernandez-Gomez et al. [165]. Although the authors do not develop their own hydrogen production model, they do present a series of equations that govern the operations of a PEM electrolyzer. These equations can help other researchers to develop HPD models.

Multi-objective problems

Multi-objective problems generally require solving for a representative set of Pareto optimal solutions and choosing between them based on the preference of the decision-maker [166].

The most commonly used method within the literature is the ε-constraint method which involves minimizing one objective function while including the other objective functions in the model as constraints bounded by a level ε [166]. If a decision maker has a pre-established preference, then a specified ε value is given, and one solution is found. However, in most cases, the model is run for various ε values. Table 7 provides a list of the methods for solving multi-objective problems identified in the literature reviewed.

Guillen-Gosalbez et al. [57] use the ε-constraint method to find trade-off solutions between minimizing total system costs and maximizing LCI for the hydrogen supply chain. To speed up the calculation of the Pareto solutions, they introduced a bi-level decomposition algorithm to break their problem into a master and subproblem. In their master problem, the integer variables representing production plants and storage facilities are removed, and variables representing transmission are relaxed. The master problem selects a subset of manufacturing, storage, and transport technologies for

their HSC. The master problem is inherently smaller than the original as it has fewer integer variables and only selects a subset of processes. In the subproblem (the planning problem), one solves for the selected set of technologies which yields an upper bound for the total cost of the system.

Hwangbo et al. [64] develop a multi-objective model for the design of a hydrogen supply chain in the Gyeongsang-do Province, South Korea, based on the steam reformation of biomethane under uncertain hydrogen demand conditions. They solve their model for each ε using the commercial solver CPLEX [167].

Sabio et al. [102] also solve a stochastic multi-objective model for the design of a hydrogen supply chain in Spain. They exploit the structure of their ε-constraint model formulation to decompose the problem into what they call a master and slave problem to expedite solving for each Pareto solution.

The weighted constraint method combines all objective functions into one composite function using weighted coefficients (the sum of which equals 1) for each function. The solution to the multi-objective problem depends strongly on the weighting coefficient. Dagdougui et al. [46] use the weighted sum method to create a single objective function designed to maximize the satisfaction of hydrogen demand from refueling stations and the industrial market, optimize the energy sold to the power network and maximize the amount of hydrogen stored in a hydrogen tank.

Lexicographic methods involve arranging the objective functions in order of importance and solving a sequence of single-objective problems, each with an additional constraint defining an upper limit using the optimal value of the previous optimization problem [166]. De-Leon Almaraz et al. [48,49] apply the lexicographic method followed by the ε-constraint method. However, no papers use lexicographic methods alone to solve a multi-objective problem.

Genetic algorithms are metaheuristic methods that use populations of candidate solutions to the optimization problem and evolve towards better solutions [166]. A genetic algorithm does not guarantee optimality and may require several runs to obtain the best solution [168]. Two papers [94,101] use a genetic algorithm, specifically a variant of non-dominating sorting genetic algorithm II to build a set of Pareto optimal solutions.

Summary and recommendations

This paper presents an extensive survey of current research topics and developments made within the scope of optimizing and modeling HSCD and HPD problems. We classified the literature according to the supply chain configuration (feedstock, production, transportation, end-users), the modeling features (decision variables, performance measures, spatial and temporal resolutions, stochasticity), and the solution approaches. We provide tables and figures that categorize the papers by the model type, spatial scale, and investment resolution, the type of performance measures used, the type of uncertainties explored, and the modeling approaches and solution methods used for various problem types (linear vs. nonlinear; single objective vs. multi-objective; deterministic vs. stochastic). We identified several gaps in the literature that should be addressed and explored further to ensure a better representation of the hydrogen supply chain.

This review shows that hydrogen demand projections used in literature are based largely on the penetrations of HFCVs. A holistic view of the hydrogen demand projections for various sectors is necessary to obtain more realistic results for the configuration of the hydrogen economy. Hydrogen can help decarbonize the transportation and industrial sectors, while providing added flexibility to the power sector. Therefore, hydrogen supply chain models can receive exogenous demand from other industries or be optimized using an integrated approach to allow additional endogenous demand requirements.

Integrated sector modeling will not only offer endogenous hydrogen demands but it allows one sector to react to changes or perturbations in another. One sector can generate the prices and availability factors for its given commodities, to which another sector, say the hydrogen sector, can respond, for instance by determining the type of production technologies to build or operate. Of great importance are co-optimizing or integrating hydrogen supply chains with its feedstock systems. It will be important to consider water, natural gas, biomass, electricity, and critical material supply chains to ensure that hydrogen models have a realistic representation of the feedstocks available to produce the commodity.

On a similar note, hydrogen supply chain models can be integrated with computable general equilibrium (CGE) models. CGE models fit economic data to a set of equations that capture the general structure of the economy and the behavioral response of firms, households and the government. These models capture the entire economy and consider interactions between different areas. By coupling energy system models with CGE, one can assess the role a new sector like hydrogen can have on the overall economy. In the same respect, changes in the economy determined by CGE models can be fed to energy system models, namely to assess the best investment opportunities and appropriate recovery factors.

Regarding spatial scales, most of the literature used a national or regional scope. Some generic imports are considered in some papers; however, taking an international perspective will allow decision-makers to more accurately model trading dynamics and consider geopolitical climates. International trading of hydrogen will similarly integrate the addition of other transport mechanisms such as shipping vessels. In

Table 7 – Solution methods for multi-objective problems identified in the literature reviewed.

Weighted sum	Lexicographic	ε-constraint	Fuzzy	Genetic Algorithm	Lexicographic + ε-constraint	Not Specified
46		38, 39, 50, 57, 64, 77, 92, 93, 96, 97, 102, 103, 118	53, 60	94, 101	48, 49	63, 79, 83

particular, additional transport mediums should be explored, given the difficulty of storing and transporting hydrogen. More research should focus on the long-haul transport of hydrogen as ammonia or LOHC and the cost associated with such transport.

Another aspect that requires additional attention is the temporal resolution of the operational decisions along the hydrogen supply chain. Most literature currently includes the average daily resource availability and hydrogen demand. Coarse time resolutions ignore correlations between potential feedstocks, for instance the availability of solar and wind may be complementary in some parts of the word. Thus, coarse resolutions may not fully capture variability and extremes in supply and demand which may result in suboptimal systems that build excess or insufficient production capacity to meet real-world hydrogen demand. Needed are models with a high temporal resolution to better capture the variability of renewable resources such as wind and solar and integrate them with hydrogen production. Finer operational resolutions will provide better feedback for investment decisions. Similarly, the application of clustering techniques to compress weather historical data from years into a number of representative days leads to the reduction of weather variability and extreme events from the input data. Underestimating variability and extreme events may have an impact on the design hydrogen supply chains, and therefore the utilization of clustering techniques requires further investigation.

The impact of climate change on the hydrogen supply chain and production are largely neglected in the literature. Changes in weather patterns in different regions will generally impact on the feedstock availability. Higher temperatures can reduce the efficiency and power output PV, and alternation in wind patterns may limit the amount of wind power that can be generated in a region. Changes in rainfall and temperature can also shift the availability of biomass resources. Furthermore, as climate change becomes more pronounced, it is expected that extreme days become harsher and more frequent. Therefore, it is important to take such changes into consideration so as to ensure that a resilient supply chain configuration is built.

Throughout the review we mentioned that leakage is important to consider in the production and supply chain of hydrogen. Hydrogen has an indirect global warming potential and therefore it should be included in GWP metrics but also in the cost metric so to prevent leaks via retrofits, and new technology.

Realistic models are necessary to get a better representation of real-world settings to capture the uncertainty of future projections, whether it be hydrogen demand, technology production capital costs, or resource availability to name a few. Realistic models will require a large number of scenarios which may render them intractable when using many of the methods found in the literature. Decomposition methods such as those suggested by Guillen-Gosalbez et al. [57] and Sabio et al. [102], will prove effective in breaking down the problems into smaller, easier-to-solve components.

The development of a hydrogen economy is being pushed by numerous countries and institutions worldwide. However, with any new development come financial risks and uncertainties. Models for the HSCD problems should therefore consider the prospect of having financial losses or stranded assets. In other respects, the uncertainty of system resiliency and reliability should also be considered.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijhydene.2022.12.242.

Index of Abbreviations

CCR	continuous catalytic reforming
CCS	carbon capture and storage
DP	dynamic program
DRM	dry reformation of methane
EMC	emission mitigation cost
GWP	global warming potential
HFCV	hydrogen fuel cell vehicle
HSCD	hydrogen supply chain design
HPD	hydrogen process design
I	international
LCI	life cycle indicator
LOCH	liquid organic hydrogen carrier
LP	linear program
MILP	mixed integer linear program
MINLP	mixed integer nonlinear program
MPC	model predictive control
N	national
NLP	nonlinear program
MPC	model predictive control
OPEX	operational expenditures
R	regional
SIM	simulation
SMR	steam methane reformation
TEA	techno-economic analysis
U	urban

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